Sanchit Sharma

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Education

Maharaja Surajmal Institute of Technology (GGSIPU), New Delhi — *Sept 2023 – Present*
B.Tech – Electronics & Communication Engineering **CGPA: 9.123 / 10** — *(Expected June 2027)*

Technical Skills

Languages: Java, Python, JavaScript, SQL, HTML/CSS
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOPs), Operating Systems, DBMS, Computer Networks, Time/Space Complexity Analysis
Databases: MySQL (relational), MongoDB (NoSQL)
Frameworks & Tools: Node.js, Express.js, React.js, REST APIs, Git/GitHub, Scikit-learn, Pandas, NumPy

Experience

Research Intern — *New Delhi*
DRDO, Ministry of Defence, Govt. of India — *June 2025 – July 2025*

- Designed and implemented **performance-critical algorithms in Java** for defence systems analysis at ISSA, applying Big-O complexity analysis to optimize for correctness and runtime efficiency under strict reliability constraints.
- Translated ambiguous, broadly-defined mathematical and domain-specific requirements into clean, modular, maintainable code — operating in a high-stakes environment with zero tolerance for failure.

Teaching Assistant, DSA & Web Development — *Remote*
Apna College — *June 2025 – Nov 2025*

- Mentored **5,000+ students** on data structures, algorithms, and full-stack web development; resolved 100+ technical doubts weekly, consistently reinforcing problem-solving from first principles and time/space trade-off analysis.

Projects

Finovate — Full-Stack FinTech Application | React.js, Node.js, Express.js, MongoDB, REST APIs — *Live Demo*

- Built a scalable full-stack financial platform with real-time stock charts, secure user authentication, dummy trade execution, and AI-powered market insights serving concurrent users.
- Architected a three-tier system — React front-end, RESTful Node.js/Express.js service layer, and a MongoDB data layer — designed as decoupled, independently scalable components.

RAG Pipeline — Retrieval-Augmented Generation | Python, LangChain, ChromaDB, Gemini API — *GitHub*

- Built an end-to-end RAG pipeline (LangChain, ChromaDB) implementing multiple chunking strategies (semantic, agentic) and retrieval techniques — multi-query retrieval, reciprocal rank fusion, hybrid search, and reranking — to improve answer relevance.
- Swapped the default OpenAI stack for Google's Gemini API for generation, and added history-aware retrieval for multi-turn conversational Q&A.

SmartAttend — AI Attendance System | Python, dlib, SVM, Resemblyzer, Librosa, Supabase — *GitHub*

- Built a dual-mode biometric attendance system: 128-d facial embeddings (dlib) classified via SVM, and speaker verification using Resemblyzer voice embeddings with Librosa-based audio segmentation for bulk voice roll calls.
- Designed a Supabase (PostgreSQL) backend with bcrypt-secured teacher/student roles and QR-code-based self-enrollment; building out the full pipeline as a Streamlit app.

Research & Achievements

Published Research Paper — NCI-TIDE 2025: Conducted independent research, authored and presented a peer-reviewed paper at a national-level conference. — *December 2024*
DSA with Java Certification — Apna College: Advanced data structures, algorithms, graph theory, dynamic programming, and competitive problem-solving in Java. — *October 2025*
Full Stack Web Development — Apna College: End-to-end MERN stack application development and deployment. — *December 2024*
Microsoft Student Chapter — Web Developer, MSIT — *Sept 2024 – May 2025*
Student Representative, Grievance Redressal Committee, MSIT — *Aug 2025 – Present*